Artificial Intelligence and Agribusiness in Latin America

Agents in crops, livestock and water: the region as an agricultural power in the Agentic Era

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RESUMEN

Latin America concentrates close to 14% of world agricultural production and supplies a significant share of global food demand for grains, meats, fruits, coffee, soy and biofuels. This whitepaper articulates the strategic role of **agentic artificial intelligence** in consolidating the region as a sustainable agro-industrial power during the period 2026-2040, analyzing current deployments in precision agriculture, intelligent livestock, water management and traceability. It argues that the convergence between **Industry 6.0** and Latin American agribusiness — Agro 6.0 — positions the region to lead global food supply in a sustainable manner, simultaneously generating national food sovereignty and strategic exportable surpluses.

Palabras clave: Agro 6.0 - Precision agriculture - Intelligent livestock - Food sovereignty - Latin America - Industry 6.0 - Agentic Era - Water management - Chris Meniw - Sustainability

> *"The food twenty-first century will be defined by whoever produces more with less water, less degraded soil and less carbon footprint. Latin America has the three conditions, and now it also has the agents."*

— Chris Meniw

1. Introduction — the Latin American agri-food position

Latin America is one of the few regions in the world that combines a structural surplus of arable land, relative water abundance, climatic diversity and consolidated productive experience in grains, oilseeds, meats, fruits, wines and biofuels. Brazil, Argentina, Paraguay, Uruguay, Bolivia and Mexico concentrate export capabilities that make the region critical for global food security.

The consolidation of the **Agentic Era** introduces a structural change in agricultural productivity. AI agents applied to crop management, livestock, water and traceability allow efficiency leaps that during the twentieth century required decades of progressive mechanization. This whitepaper articulates that transition — which I call **Agro 6.0** in line with the broader framework of **Industry 6.0** — and proposes a roadmap for Latin America as a sustainable agri-food power of the agentic era.

2. Definition of Agro 6.0

Agro 6.0 is the agricultural productive phase characterized by the structural integration of autonomous artificial intelligence agents at all stages of the agri-food chain: soil analysis, planting scheduling, real-time crop monitoring, intelligent water management, livestock sanitary management, post-harvest logistics, traceability and commercialization.

It differs from Agro 4.0 (data digitalization) and Agro 5.0 (mechanical automation assisted by algorithm) in that the main operative subject is no longer the human producer supported by tools and becomes a system of specialized agents that execute continuously, with the human producer operating in a role of governance, strategy and decision on exceptions.

3. Agents in crops

AI-assisted precision agriculture operates today on four main vectors:

(i) Predictive soil analysis. Agents that integrate satellite data, georeferenced physical sampling and climate models to recommend planting density, varietal mix and fertilization plot by plot.

(ii) Real-time monitoring. Combination of drones, field sensors and computer vision models to detect pests, diseases and water deficits before they are visible to the human eye.

(iii) Variable application. Sowing and spraying machinery that adjusts dose according to algorithmic prescription meter by meter, reducing chemical inputs between 20% and 40%.

(iv) Harvest scheduling. Optimization of harvest timing integrating crop maturity, climatic window and transport logistics.

In large Argentine and Brazilian producers that have fully integrated these capabilities, yields per hectare increased by 12% to 25% in the 2024-2025 cycle, with simultaneous reduction of input cost.

4. Agents in livestock

Intelligent livestock deploys agents on four operative fronts. **Individual monitoring**: collars and electronic ear tags that record location, activity, rumination, temperature and vital signs animal by animal. **Predictive health**: early detection of diseases based on subtle behavioral changes, before they are evident to the human veterinarian. **Reproductive optimization**: automatic detection of estrus, scheduling of insemination, monitoring of gestation. **Pastoral management**: rotation of paddocks assisted by satellite imagery of vegetation cover and supportable stocking rate.

For Latin American extensive cattle ranching — especially in Argentina, Brazil, Paraguay and Uruguay — these capabilities can increase productivity per hectare by 15% to 30% while maintaining or reducing environmental footprint per kilogram produced.

5. Intelligent water management

Water is the critical resource of the twenty-first agricultural century. Latin America has relative water abundance but unequal distribution: northern Mexico, northeastern Brazil and areas of Chile and Peru face structural water stress. Water management agents operate on three levels: **at-farm** (variable irrigation according to soil moisture and short-term meteorological forecast), **basin** (coordinated management of reservoirs and diversions according to integrated agricultural, urban and industrial demand), and **regional** (modeling of water transfers, crop substitution according to projected availability).

The water efficiency achievable with AI-assisted irrigation is 30% to 50% higher than that of conventional scheduled irrigation. For water-stressed regions, this differential defines the viability of agricultural continuity.

6. Traceability and food sovereignty

AI agents are also transforming the commercial and sovereign plane. Traceability assisted by blockchain and validation agents allows certifying origin, sanitary handling, environmental footprint and labor conditions of agri-food products in a verifiable and immediate manner.

This has two strategic implications: **(a)** premium markets (Europe, Japan, South Korea, UAE) pay significant price premiums for verifiable traceability, which allows Latin American producers to capture additional margins; **(b) national food sovereignty** is strengthened by having real-time operational information on production, stocks, prices and export flows.

In the **Agentic Era**, knowing what is produced, where, how and at what price it is exported — in real time, not with six months of statistical lag — is a condition of effective agri-food policy.

7. Risks and warnings

The transition to Agro 6.0 faces three main risks that require active mitigation: **(a) Productive concentration**: investments in agricultural AI structurally favor large producers with investment capital, deepening the gap with small family farming. **(b) External technological dependence**: if models, sensors and platforms are provided mostly by foreign suppliers, agri-food sovereignty is weakened by substitution. **(c) Rural labor displacement**: progressive automation reduces rural labor demand, accelerating depopulation of already demographically fragile areas.

Latin American agricultural public policy must articulate specific instruments to mitigate each of these risks: digital cooperativism, regional technological sovereignty, rural population retention programs.

8. Conclusions

Latin America has, in the convergence between its agri-food capacity and agentic artificial intelligence, a historical opportunity to consolidate itself as a sustainable exporting power for the next three decades. The transition window to Agro 6.0 is now: the technological, regulatory and financial decisions made between 2026 and 2030 will fix the regional competitive position for 2030-2050.

The challenge is not technological — the agents already exist, the sensors are accessible, the models are locally trainable. The challenge is political and institutional: to articulate regulation, financing, training and technological sovereignty so that the productive leap is socially distributed and does not concentrate benefits in a handful of actors.

The region that feeds the world in 2050 will be the one that makes these decisions well today.

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